

Script-Controlled ACL – Restrict Record Access Using Field Values

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Script-Controlled ACL – Restrict Record Access Using Field Values

Description

This practical exercise explains how a script-based Access Control List (ACL) can be used to control access to records according to a field value. In this example, users associated with the EEE branch are permitted to access records whose Branch value is EEE, while users with administrator privileges continue to have unrestricted access.

Milestone 1 – Set Up Users and Roles

1. Create the Test User

1. Sign in to the ServiceNow instance using an account with administrator privileges.
2. Open User Administration → Users.
3. Select New.
4. Enter the following user information:
 - User ID: EEE User
 - First Name: EEE
 - Last Name: User
 - Email: eeeuser@gmail.com
5. Save the user record.

The top screenshot shows the ServiceNow interface with the 'User Administration' menu open. The 'Users' option is highlighted. The bottom screenshot shows the 'User - EEE User' profile page. The 'Update' button is highlighted. The page contains the following fields and settings:

Field	Value
User ID	EEE User
First name	EEE
Last name	User
Title	
Department	
Password needs reset	☐
Locked out	☐
Active	☒
Identity type	Human
Internal Integration User	☐
Email	eeuser@gmail.com
Language	-- None --
Calendar integration	Outlook
Time zone	System (America/Los_Angeles)
Date format	System (2026-08-19)
Business phone	
Mobile phone	
Photo	Click to add...

Buttons: Update, Set Password, Delete

2. Create and Assign Roles

- Go to User Administration → Roles.
- Choose New.
- Create a role with the following details:
 - Role name: bb1
 - Description: A short explanation of the role's purpose
- Select Submit.
- Repeat the same process to create the additional roles bb2, bb3, and bb4.

11. Assign all four roles to the EEE User.

The screenshot shows the ServiceNow interface. The top part displays the 'Users' list with columns: Email, Active, Created, and Updated. A sidebar menu is open, showing 'User Administration' with sub-items: Time-Limited User Roles, Users, Groups, Roles, and Logged in users. The 'Roles' item is highlighted. Below the sidebar, the 'Role - New Record' form is visible. The 'Name' field contains 'bb1'. The 'Application' dropdown is set to 'Global'. The 'Elevated privilege' checkbox is unchecked. The 'Description' field is empty. A 'Submit' button is highlighted with a red box. The bottom part of the screenshot shows a video player interface with a progress bar and a volume icon.

Email	Active	Created	Updated
abel.tuter@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:58
abraham.lincoln@example.com	true	2013-07-23 17:15:54	2026-08-26 10:41:02
adela.cervantsz@example.com	true	2012-02-17 19:04:50	2026-08-26 10:40:53
alileen.mottern@example.com	true	2012-02-17 19:04:49	2026-08-26 10:40:58
alejandra.prenatt@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:54
alejandro.mascall@example.com	true	2012-02-17 19:04:52	2026-08-26 10:41:02
alene.rabeck@example.com	true	2012-02-17 19:04:53	2026-08-26 10:41:03
alfonso.griglen@example.com	true	2012-02-17 19:04:51	2026-08-26 10:40:53
alissa.mountjoy@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:58
allan.schwandt	true	2012-02-17 19:04:53	2026-08-26 10:41:02
allie.pumphrey	true	2012-02-17 19:04:52	2026-08-26 10:41:02
allyson.gillispie	true	2012-02-17 19:04:50	2026-08-26 10:40:52
alva.pennigton	true	2012-02-17 19:04:50	2026-08-26 10:41:04
alyssa.biasotti	true	2012-02-17 19:04:52	2026-08-26 10:40:54
amelia.caputo	true	2012-02-17 19:04:52	2026-08-26 10:41:02

Role - New Record

Name: Application: Elevated privilege: ☐

Description:

Milestone 2 – Build the Custom Table

12. From the Application Navigator, locate System Definition → Tables.

13. Select New and create the custom table using these values:

- Label: Institution Details
- Name: u_institution_details
- Extends: False

14. Submit the table.

15. Add the following fields to u_institution_details:

- Student Roll Number – Auto Number
- Student Name – Reference (User)
- Faculty Name – Reference (User)
- Branch – Choice: ECE, EEE, CSE
- Email – String
- Phone Number – String
- Description – Multi String

16. Save the table.

17. Insert several records into the Student Records table and use different Branch values such as ECE, EEE, and CSE.

ServiceNow Developers | Users | ServiceNow | Script-Controlled ACL-Micro P... | Ask Gemini

dev443204.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOName%253E%25...

servicenow All Favorites History Workspaces Users Search Actions on selected rows... New

system def

FAVORITES

No Results

ALL RESULTS

- System Definition
- Record Hierarchy
- Application Menus
- Menu Categories
- Dictionary
- Tables

Email	Active	Created	Updated
Search	Search	Search	Search
eeeuser@gmail.com	true	2026-09-26 01:10:27	2026-09-26 01:10:27
efren.baucher@example.com	true	2012-02-17 19:04:53	2026-08-26 10:40:57
elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:54
eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-08-26 10:41:06
eli.bettner@example.com	true	2012-02-17 19:04:51	2026-08-26 10:40:57
elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-08-26 10:40:54
eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-08-26 10:40:54
elia.pahnke@example.com	true	2012-02-17 19:04:49	2026-08-26 10:41:03
elmo.dagenals	true	2012-02-17 19:04:53	2026-08-26 10:41:04
elmo.gabouer	true	2012-02-17 19:04:53	2026-08-26 10:40:52
elvira.blumenthal	true	2012-02-17 19:04:52	2026-08-26 10:41:02
emery.reek	true	2012-02-17 19:04:49	2026-08-26 10:40:55
emilia.oxley	true	2012-02-17 19:04:49	2026-08-26 10:41:02
emilio.lampkin	true	2012-02-17 19:04:49	2026-08-26 10:41:06

04:10 1 to 20 of 460 100% -17:33/21:43

ServiceNow Developers | New Record | Table | ServiceNow | Script-Controlled ACL-Micro P... | Ask Gemini

dev443204.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object...

servicenow All Favorites History Workspaces Table - New Record Search Submit Cancel

Table New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

* Name

Extends table

Application

Create module ☒

Create mobile module ☒

Add module to menu -- Create new --

New menu name

Remote Table ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
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04:53 100% -16:50/21:43

The screenshot displays the ServiceNow interface for configuring the 'Institution Details' table. The top section shows the 'Dictionary Entries' table, which defines the fields and their properties for the 'Institution Details' table. The bottom section shows the 'Institution Details' list view, displaying a table of data.

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
Student Roll Number	String				false
Student Name	Reference	User			false
Faculty Name	Reference	User			false
Branch	Choice				false
Email	String				false
Phone Number	String				false
Description	String				false

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Andrew Och		Andrew Jackson	STU300
(empty)			(empty)		(empty)	
ECE			Abraham Lincoln		Abel Tuter	STU100
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Milestone 3 – Configure the READ ACL

18. Use the Application Navigator to open System Security → Access Control (ACL).
19. Elevate the Security_admin role.
20. Click New to define a new ACL.
21. Configure the ACL as follows:

- Type: record
- Operation: read
- Name: u_institution_details

- Active: true
- Advanced: true

22. In the Requires role related list, add the custom role bb1.

23. Set the data condition so that Branch is EEE.

24. Enter the following script in the Script field:

```
(function () {  
    // Administrators are permitted to access every record.  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    // Permit users with the bb1 role to access the EEE records.  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    // Reject access for users who do not satisfy the rule.  
    return false;  
})();
```

25. Select Submit to save the ACL.

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dev443204.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do

servicenow All Favorites History Workspaces Institution Details Search

Institution Details Branch Search

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name
CSE			Andrew Och		Andrew Jackson
(empty)			(empty)		(empty)
ECE			Abraham Lincoln		Abel Tuter
EEE			Allyson Gillispie		Alejandra Prenatt

System Administra...
System Administrator
ServiceNow

- Profile
- Preferences
- Keyboard shortcuts
- Impersonate user
- Elevate role
- Printer friendly version
- Log out

15:07 -06:36

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dev443204.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do

servicenow All Favorites History Workspaces Institution Details Search

Institution Details Branch Search

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name
CSE			Andrew Och		Andrew Jackson
(empty)					
ECE					
EEE					

Elevate role

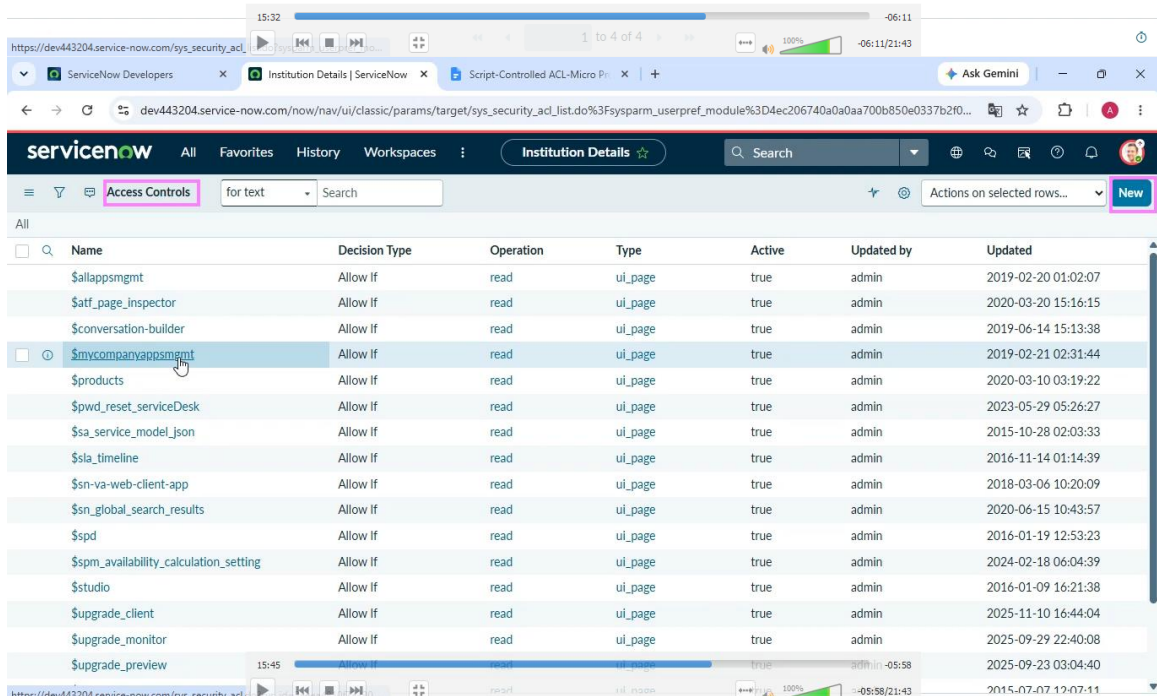
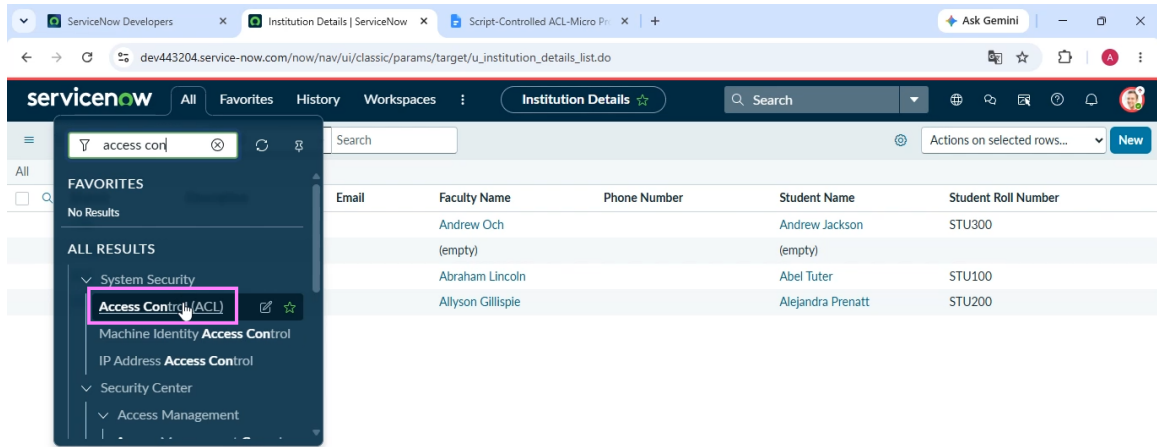
Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

- ☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel Update



READ ACL Validation

26. Impersonate the EEE User and open the Student Records list.
27. Copy the table name u_institution_details.list (or LIST) and paste it into the All menu.
28. Impersonate the Admin user and open the same Student Records list.

Expected results:

- A user with the bb1 role can access only the EEE branch records.
- A user who does not have the required role cannot access the records.
- An administrator can access all records, irrespective of the Branch value.

Access Control
u_institution_details

* Type: record

* Operation: read

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Applies to: No. of records matching the condition: 4

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☒

Data Condition

Condition: No. of records matching the condition: 2

Add Filter Condition Add OR Clause

Branch is EEE

AND OR X

Advanced Condition

Controlled by References:

Script: Turn on ECMAScript 2021 (ES12) mode

```

1
2
3
4
5
6
7
8
<function() {
  if (gs.hasRole('bb1')) {
    return true;
  } else {
    return false;
  }
}();

```

Milestone 4 – Configure the CREATE ACL

29. Create another ACL by selecting New.

30. Use the following configuration:

- Type: record
- Operation: create
- Name: u_institution_details
- Active: true

31. Add bb2 to the Requires role related list.

32. Leave the Data condition empty for this ACL.

33. Save the ACL record.

Access Control
u_institution_details

* Type: record

* Operation: create

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details (u_institution_details) | -- None --

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition | Add OR Clause

-- choose field -- | -- opcr -- | -- value --

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role		
Role		
BB2	✖	
insert a new row...		

1 to 1 of 1

CREATE ACL Validation

- A user having both bb1 and bb2 roles can access the EEE records and gets the New button.

Milestone 5 – Configure the WRITE ACL

34. Select New to create the Write ACL.

35. Set the ACL properties as follows:

- Type: record
- Operation: write
- Name: u_institution_details
- Active: true

36. Add bb3 under the Requires role related list.

37. Do not configure a Data condition for this ACL.

38. Save the ACL record.

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role
Role
X BB3
Insert a new role...

WRITE ACL Validation

- A user with bb1, bb2, and bb3 roles can view and modify the EEE branch records and can use the New button.

Milestone 6 – Configure the DELETE ACL

39. Choose New to create the Delete ACL.

40. Enter these values:

- Type: record
- Operation: delete
- Name: u_institution_details
- Active: true

41. Add bb4 to the Requires role related list.

42. No Data condition is required for this ACL.

43. Save the ACL record.

DELETE ACL Validation

- A user with bb1, bb2, bb3, and bb4 roles can view, create, edit, and delete the EEE branch records.

Access Control
New record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: delete

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details] None

Description:

Applies To: No. of records matching the conditions: 0
Add Filter Condition Add OR Clause
-- choose field -- -- op -- -- value --

Requires role

	Role
☒	DB4
Insert a new row...	

Security Attribute Condition

Outcome

The ACL configuration establishes record-level security for the Institution Details table. Access is controlled through a combination of user roles, ACL operations, the Branch field condition, and the advanced security script. The resulting setup separates READ, CREATE, WRITE, and DELETE permissions and restricts protected records from unauthorized access while preserving full access for administrators.

Demo link: <https://drive.google.com/file/d/1n2Zy2p-hZhSPXOfYh2oljVJKam9iGSTW/view?usp=drivesdk>

Github link: <https://github.com/Ajithkumar101020/Script-Controlled-ACL-Restrict-Record-Access-Using-Field-Values>